Sidharth Sajith

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Professional Summary

Robotics and Automation engineering student focused on robotics software, computer vision, embedded systems, and edge AI. Hands-on experience developing Python-based algorithms, ROS 2 communication pipelines, OpenCV applications, and IoT systems. Interested in autonomous systems, intelligent perception, and deploying software across resource-constrained hardware.

Technical Skills

- **Programming:** Python, C, C++, JavaScript, SQL
- **AI / Machine Learning:** PyTorch, TensorFlow, Scikit-Learn, OpenCV, MediaPipe, YOLOv8, YOLO26, TFLite, Hugging Face, Gemini API, RAG, AWS Rekognition
- **Robotics:** ROS 2 Humble, Nav2, Gazebo, NVIDIA Isaac Sim, URDF, SLAM, OctoMap, Visual-Inertial Odometry, Bayesian Search
- **Embedded / IoT:** ESP32, NodeMCU, Arduino, ESP-IDF, PlatformIO, MQTT, I2C, SPI, PCB Design
- **Development & Tools:** Linux, Git, Docker, Flask, React.js, Streamlit, MySQL, REST APIs, WebSockets, Qualcomm AI Hub, ARCore

Experience

AI & IoT Solutions Intern (Part-time) *Oct 2025 – Present*
Innovizta (Campus Startup), Kottayam, Kerala

- Developed **NeuralHealth**, a multithreaded desktop application integrating MediaPipe-based posture tracking, TensorFlow emotion recognition, and ESP32 environmental sensor data
- Built **Breathe+**, an IoT-enabled smart inhaler system with MQTT telemetry, a Streamlit monitoring dashboard, and automated Telegram alerts for dosage and air-quality monitoring
- Developed an intelligent document automation pipeline using YOLOv8 and the Gemini API to extract and cross-verify purchase orders, delivery records, and invoices

Embedded Systems Research Intern (Part-time) *Aug 2025 – Present*
ICCC Saintgits (Taiwan–India), Kottayam, Kerala

- Engineered a six-node distributed environmental sensor network using ESP32-based NodeMCU boards with AHT10, HL83, and SDS011 sensors and MQTT-based data transmission
- Integrated Open-Meteo and OGD India REST APIs using multithreaded Python to synchronize external reference data with real-time physical sensor streams
- Developed backend data-processing scripts to parse MQTT payloads and maintain structured MySQL records for historical environmental analysis

R&D Intern (Research Project) *Aug 2025 – Present*
Kerala Electrical & Allied Engineering Co. Ltd. (KEL), Mamala, Kerala

- Developed an OpenCV-based computer vision component to track transformer core laminations during stacking operations, exploring vision-assisted monitoring for industrial automation
- Conducted feasibility research on automated robotic systems for stacking transformer core laminations in industrial manufacturing environments
- Evaluated sensing requirements and automation concepts for integration with existing industrial workflows and safety systems

Projects

Embodied Semantic Asset Recall Robot *Ongoing*
Python, ROS 2, Bayesian Search, Android, WebSockets

- Developing the software for an offline semantic object-retrieval robot, focused on persistent object memory and probabilistic search for relocated or missing assets

- Implemented a Bayesian active-search module with a discrete probability grid, sensor model, and next-best-view planning; achieved 12/12 passing automated tests
- Contributed to the Android sensing application and WebSocket-to-ROS 2 communication bridge; the URDF software model loads in RViz for robotics development

Autonomous Vehicle Research Platform (ADAS) *Ongoing*
PyTorch, TFLite, Qualcomm AI Hub, Snapdragon 8 Gen 2, NVIDIA Isaac Sim

- Architected a heterogeneous edge-AI pipeline for a Snapdragon 8 Gen 2 platform, distributing perception workloads across GPU, NPU, and CPU resources
- Developed PyTorch tracing and ONNX model-conversion workflows for deploying computer vision models through Qualcomm AI Hub
- Diagnosed edge deployment challenges involving tensor precision, static input dimensions, delegate compatibility, memory utilization, and thermal constraints

EduBot: Interactive Educational Robot *Jun 2026*
Raspberry Pi 5, AWS Rekognition, LLM, Speech AI

- Developed an AI-powered educational robot on Raspberry Pi 5 with integrated large language model, speech-to-text, and text-to-speech capabilities
- Integrated AWS Rekognition for vision-based student identification and developed a custom administration interface
- Designed bilingual English and Malayalam conversational workflows for interactive primary-school learning

Aries Radar System *Apr 2025*
Flask, MediaPipe, WebSockets, JavaScript

- Migrated a desktop-based hand-tracking prototype to a Flask and JavaScript web application using WebSockets for real-time browser-based visualization

Achievements & Leadership

- **National Top 50**, e-Yantra Robotics Competition 2025
- **Chairperson**, The Robotics Society of India (RSI), Saintgits Chapter
- **Online Workshop**, RSI Saintgits Chapter – conducted an online OpenCV workshop for approximately 40 participants, covering practical image processing and computer vision concepts
- **Hardware & Circuit Lead**, Srishti 2025 Hackathon – led a five-member team in developing multiple functional IoT and embedded systems prototypes within four days
- **Core Executive Committee Member**, Saintgits Engineering Students Association (SESA)
- **Autonomous Systems Volunteer**, BAJA SAE – trained YOLO-based perception models for traffic signs, signals, and object detection for autonomous vehicle applications
- **Participant**, AI Impact Summit Buildathon, HCL GUVI – Mar 2026

Education

Bachelor of Technology in Robotics and Automation *Expected Jul 2027*
Saintgits College of Engineering (Autonomous), Kottayam, Kerala